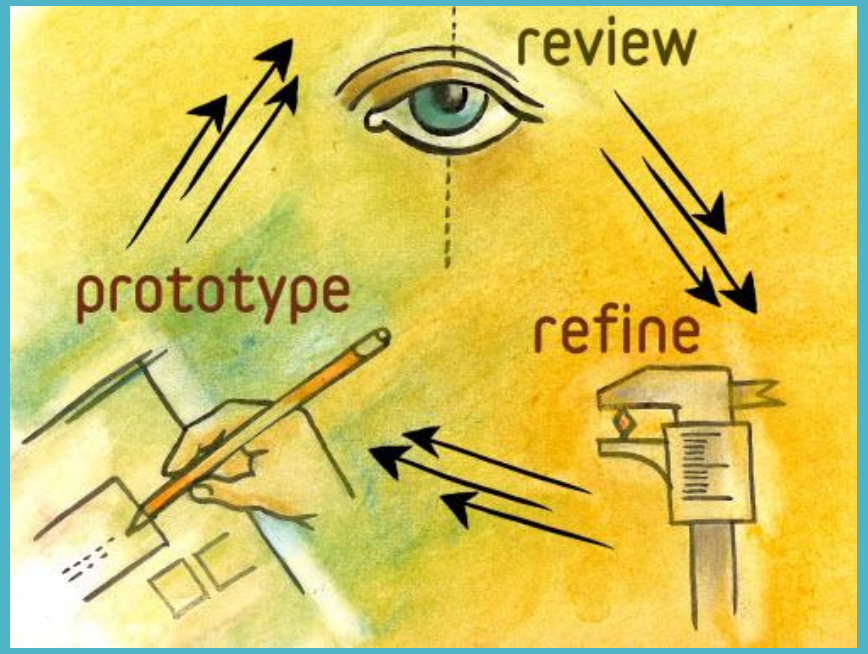
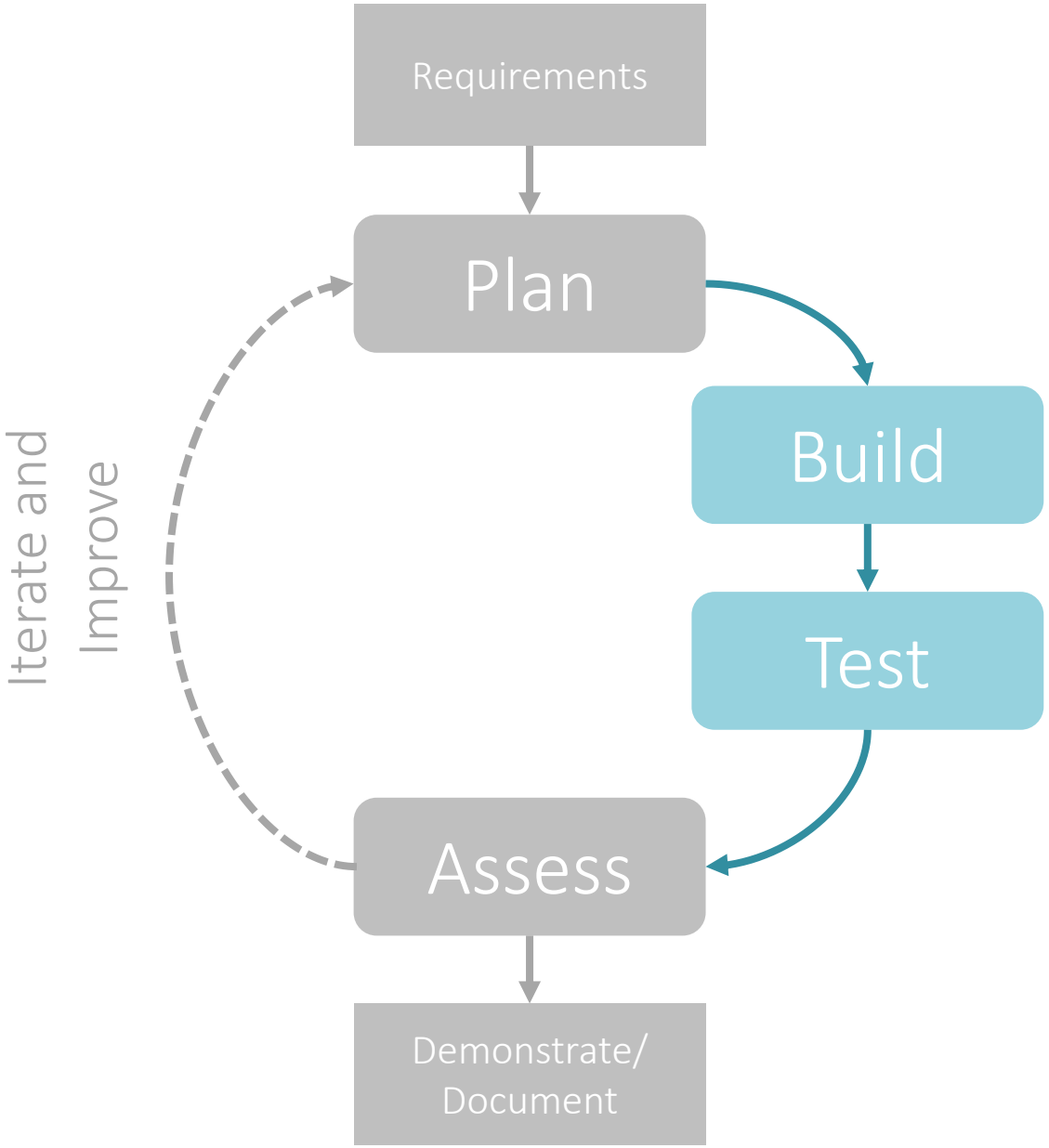
ECE115 Fast Prototyping

ECE115
Michael Yip
UC San Diego



What do we mean by *Fast*?

- As fast as possible, try out an idea
- Plan with failure in mind
- Test intelligently
- Avoid the feeling of “sunken cost” at the cost of a prototype’s quality



ECE115 Teaching Team

Instructors

Michael Yip – yip@ucsd.edu

TAs

Zoe Samuels

Omar Hernandez

Elizabeth Peiros

Piazza

We will exclusively use Piazza for communication

All questions will be answered here in the fastest possible time.

Course Information

This is a project-based course.

Occasional lecture material

Majority of the time is for you to work on your project and interact with the teaching staff.

When you come into lab, you may start working immediately.

Time: 11am-2pm Tu/Th

Do not double-book this course! We do grading in-class and participation is being evaluated throughout the course.

Course Information

Canvas

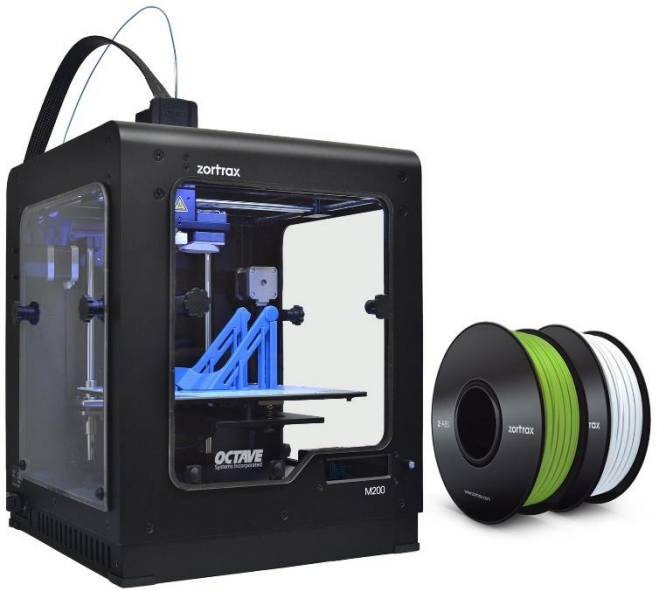
- **Find** all documentation online
  - *modules will be released the demo day of the preceding module*
- **Submit** files under Assignments
  - *Each individual needs to upload a set of files, as we will grade separately*

Questions/Discussion

- **Piazza**: rapid responses from TAs, Instructor, and fellow classmates
- **Email**: slowest method, should only be used for private matters

Future schedule conflicts: notify teammate and teaching staff immediately

MECHANICAL SKILLSET



CAD

3D modeling in Solidworks

Fabrication

Laser cutting

3D printing

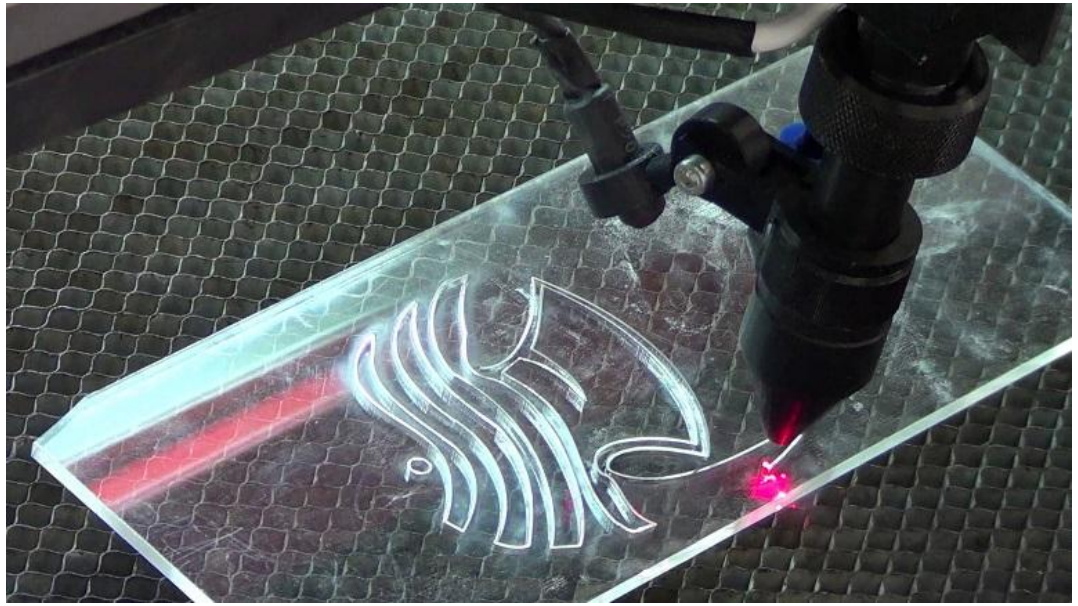
Assembly techniques

Design

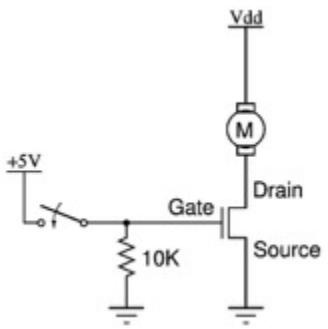
Smart Mechanical Design

Interesting mechanisms

Power and transmission

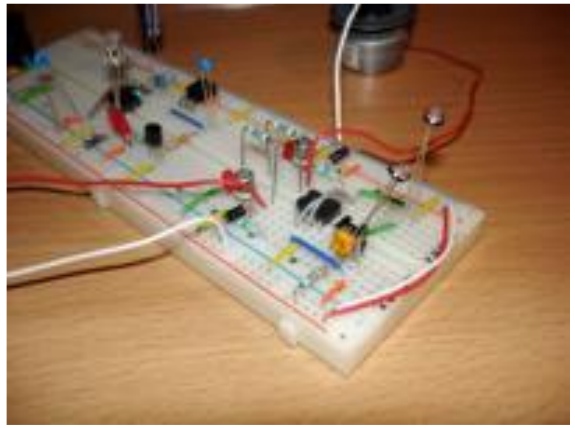


ELECTRICAL SKILLSET



Actuator and Sensor Electronics Design

- Analog and digital power electronics
- Analog Filters
- Circuit analysis and signal processing
- Microcontrollers (Arduino)



Fabrication

- Breadboarding, limited soldering

Testing

- Oscilloscopes, Multimeters, and Arduinos

What you will accomplish

At the end of the quarter, you and your lab mate have built your own pinball machine you can take home

You will understand design cycles and terminology

You will understand fast prototyping, and designing in a modular mindset

Calendar

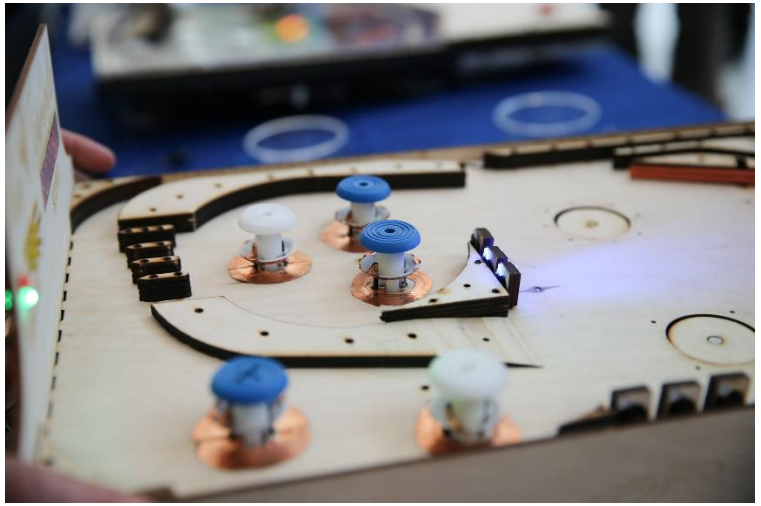
Check Syllabus

Last notes

Decide immediately (this week!) if you have the capacity to spend many hours weekly on this course.

Assignment 0: Complete the FINAid assignment.

Last Years



Note on Safety

Complete the Safety Lectures:

https://ece.ucsd.edu/makerspace

Culture of Safety

Safety first

Prevention

Questions to ask yourself:

Could this be dangerous?

What can go wrong will eventually go wrong.

Don't know how to use something? ALWAYS ask.